<table>
<tr><td>LESSON PLAN</td><td>COURSE: MDM4U</td></tr>
<tr><td>Date:</td><td>Unit of Study: Unit 1</td></tr>
<tr><td>Materials/Resources:
Mathmetics of data management study guide</td><td>Lesson Focus/Theme:
Chapter 7: Probability Distribution</td></tr>
<tr><td colspan="2">Prior Knowledge:</td></tr>
<tr><td colspan="2">Overall Expectations:

By the end of this lesson, students will:

demonstrate an understanding of continuous probability distributions, make connections to discrete probability distributions, determine standard deviations, describe key features of the normal distribution, and solve related problems from a variety of applications.</td></tr>
<tr><td colspan="2">Specific Expectations:
2.5 recognize that theoretical probability for a continuous random variable is determined over a range of values (e.g., the probability that the life of a lightbulb is between 90 hours and 115 hours), that the probability that a continuous random variable takes any single value is zero, and that the probabilities of ranges of values form the probability distribution associated with the random variable

2.6 recognize that the normal distribution is commonly used to model the frequency and probability distributions of continuous random variables, describe some properties of the normal distribution (e.g., the curve has a central peak; the curve is symmetric about the mean; the mean and median are equal; approximately 68% of the data values are within one standard deviation of the mean and approximately 95% of the data values are within two standard deviations of the mean), and recognize and describe situations that can be modelled using the normal distribution

2.7 make connections, through investigation using dynamic statistical software, between the normal distribution and the binomial and hypergeometric distributions for increasing numbers of trials of the discrete distributions (e.g., recognizing that the shape of the hypergeometric distribution of the number of males on a 4-person committee selected from a group of people more closely resembles the shape of a normal distribution as the size of the group from which the committee was drawn increases)

2.8 recognize a z-score as the positive or negative number of standard deviations from the mean to a value of the continuous random variable, and solve probability problems involving normal distributions using a variety of tools and strategies (e.g., calculating a z-score and reading a probability from a table; using technology to determine a probability), including problems arising from real-world applications</td></tr>
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Learning Goals:

By the end of this lesson, students will know the geometric distribution and hypergeometric distribution

Success Criteria:

By the end of this lesson, I will be able to know the geometric distribution

By the end of this lesson, I will be able to know the hypergeometric distribution

Assessment Strategies: Tests, quiz and project

Time: **10 min**	**Introduction/Minds-On:** Answer questions students don’t understand from last class if any
Time: **100 min**	**Lesson:** Chapter 7: Probability distribution 7.3 Geometric distribution 7.4 Hypergeometric distribution
Time: 10 min	**Extension / Wrap-up:** Review the key concepts and formula for this lesson

Notes / Reminders / Homework:
Do test #3 before or after the class in classroom for one hour

Lesson Reflection: